Examiner reports

Q1.

The majority of students correctly chose option 1, but a significant minority chose option 2 having mixed up the requirement to find σ rather than s.

Q2.

Overall students performed quite well on this question, but there were a significant minority who appeared insufficiently prepared for this topic and unable to gain much credit. In general, students performed well on parts (a) and (b), with the majority scoring full marks. Most students did not score well on part (c), failing to appreciate the significance of likely negative values within the 3 standard deviation spread. Part (d) had a very mixed response. Many students scored full marks but many others could not use the inverse normal distribution correctly.

Q5.

Most students scored the 2 marks in this question. However, a minority considered the u-values as frequencies for the corresponding v-values and so scored no marks at all.

Q6.

Many answers to this question scored no marks and suggested that such candidates, despite completing a piece of coursework, had little experience of assessing the likely accuracy of calculations. A small minority of candidates recognised that 151.5 cm was approximately 1.5 m and so Ashley's mean value was likely to be correct. However, only the few very best candidates realised that the standard deviation would be $\frac{2\times 10}{4}$ ≈ 5 cm (a property of a normal distribution) so that 26.6 cm was likely to be incorrect. Common worthless attempts were to simply compare 26.6 cm with 10 cm or to attempt a standardisation of 151.5 using 150 and 26.6 or 10.

Q7.

Most students were able to make a good attempt at this first question. In part (a), most students obtained the correct answer, but a small minority evaluated $\frac{\sum x}{n}$ for no marks. Again in part (b), there were many sound, often correct, attempts but some students divided 5840 by 730 and saw nothing suspicious about £ 8 as Steve's mean daily takings. It was pleasing to see the many students who obtained correct answers in parts (a) & (b), and then argued correctly in part (c) that Steve should not purchase the business on financial grounds.

Q8.

Part (a) caused few students any problems. A small minority stated '34' for the mode whilst, in part (a)(ii), a few others attempted to treat the data as continuous or calculated the interquartile range by using (132 – 44 = 88 → 22). As expected, except for a fairly common listing on the question paper of mid-points, the majority of answers to part (b) were obtained directly from calculators. However, many students obtained inaccurate or even completely incorrect answers. The inaccuracies were, in the main, due to using incorrect mid-points and / or replacing the value of 54 by 52 or even 50. Such students usually scored 2 of the 4 marks available. A minority of students ignored the frequencies and simply worked with mid-points and n = 12 so obtaining incorrect answers for no reward. More students than expected had difficulty with part (c). All that was required, for 2 follow-through marks, was to simply add $\frac{280}{175}$ = 1.6 to the mean value in part (b). About 1 / 3rd of students did this correctly. Of the remaining 2/3rds, about 1/2 made no attempt and a similar proportion attempted a variety of methods. Many such methods were invalid (eg [(mean from (b)) × 1.6] or [($\sum fx$ + 280)/176]) but some were valid although lengthy and often inaccurate (eg [(recalculated$\sum fx$ + 280)/175]) and so usually scored at most one mark.

Q9.

A reasonable number of candidates obtained the correct values for the mode, median and interquartile range. Some candidates even obtained an answer for the mean within the range permitted, but very few scored the 2 marks for the standard deviation. It was evident that most candidates were unsure as to how to treat grouped data with the result that they probably used incorrect mid-points, but the absence of any detail made this uncertain.

Of even more concern was the number of candidates who failed to use the frequencies, particularly when calculating the standard deviation. Calculating the mean and the standard deviation from a frequency distribution or from a grouped frequency distribution must be bread and butter to candidates entered for this paper. Disappointingly, most candidates apparently had no clue as to what was meant by ‘measure of spread’. In fact ‘mode’, ‘median’, ‘mean’, ‘bar chart’, ‘histogram’, etc were seen with varying frequencies. It should be noted that the standard responses of ‘mean and standard deviation’ or ‘median and interquartile range’ also scored no marks.

Q10.

Most candidates obtained the correct values for the mode, median, interquartile range and range, although some candidates treated the data as continuous. A small minority hedged their bets for the mode by quoting values of 253 and 20, and similar numbers did not state the difference between their correct values for the upper and lower quartiles or merely stated the range to be 227 to 271. Each of these three errors lost a mark. A majority of candidates obtained an answer for the mean within the range permitted, but far fewer scored the 2 marks for the standard deviation. It was evident that too many candidates were unsure as to how to treat grouped data with the result that they probably used incorrect mid-points, but the absence of any detail made this uncertain.

Of even more concern was the significant number of candidates who failed to use the frequencies, particularly when calculating the standard deviation. Calculating the mean and the standard deviation from a frequency distribution or from a grouped frequency distribution should be bread and butter to candidates entered for this paper. Disappointingly, most candidates apparently had no clue as to what was meant by ‘measure of spread’. In fact, the most common response was ‘mode’ but ‘mean’, ‘median’, ‘bar chart’, ‘histogram’, ‘box and whisker’, ‘stem and leaf’, ‘correlation’ or even ‘CLT’ were seen with varying frequencies. It should be noted that the standard responses of ‘mean and standard deviation’ or ‘median and interquartile range’ also scored no marks. However, when interquartile range alone was stated, the usual correct reason related to extreme values and, when just standard deviation was quoted, reference was made to it using all the data.

Q11.

Almost all candidates scored at least 3 marks with many achieving 5 or 6 marks. There was a marked improvement, compared with previous papers, in the calculation of the median and interquartile range, although a small minority of candidates produced an incomplete solution by not finding the difference between their correct upper and lower quartile values. Incorrect answers were usually attributable to slips or omissions, particularly in respect of a and b, in ordering the data.

Part (b)(i) was the least successfully answered part. Incorrect answers included "a and b unknown" or "All values are different". Other answers such as "A large range" were unclear as to whether this applied to the 'data range' (not accepted) or 'many different values' (accepted). Answers to parts (b)(ii) were almost invariably correct.

Q12.

Almost all candidates scored at least 4 marks, with many achieving 7 or 8 marks. This type of question has appeared fairly regularly of late, which may explain why there was a marked improvement in the calculation of the median and interquartile range. Incorrect answers were usually attributable to slips or omissions, particularly in respect of a and b, in ordering the data. A minority of weaker candidates attached the given values as frequencies to the numbers 1, 2, 3, ...,15.

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Q13.

The calculation/identification of mean, median and mode has appeared on recent past papers so the number of candidates not scoring all 5 marks in part (a) was disappointing. Indeed it was not unusual to see scores of fewer than 3 marks. Common errors were to calculate the mean as $\frac{52}{8}$ = 6.5, the median as $\frac{4+5}{2}$ = 4.5 (by disregarding the number of weeks) and to quote the modes as 1 and 13. Some candidates even found two sets of values; one for the number of births and one for the number of weeks.

Many answers to part (b) were also incorrect. Common responses were: mean as not a whole number (0 marks), median as does not use all data (0 marks); mode as does not take account of all values (1 mark). These, and other incorrect attempts, suggested that many candidates did not have a sufficient understanding of topics on Numerical Measures as detailed in the specification.

Q14.

Candidates appeared better prepared for part (a) than in the past, so their answers, usually directly from calculators, were in the main correct. It was also pleasing to note the large proportion of correct answers to parts (b)(i) and (ii). In the former, some candidates mysteriously obtained the correct answer despite totally incorrect work in part (a) and, in the latter, some answers were backed-up by a correct calculation.

In part (b)(iii), only the most able candidates were able to make valid statements as to why a normal distribution was not an appropriate model. Many invalid attempts made reference to the sample size or the Central Limit Theorem, neither of which had any relevance here. Candidates should be aware that, if X is normally distributed, then, for any sample size, the sample mean, $\bar{X}$, will be normally distributed regardless of the Central Limit Theorem; for other cases, the Central Limit Theorem states that, for sufficiently large sample size, the sample mean will be approximately normally distributed, but candidates should not infer from this that the population values, or actual sample values, will themselves be approximately normally distributed.

Q15.

Candidates appeared better prepared for part (a) than in the past but it remains a concern that far too many candidates cannot sensibly tackle this type of question. Whilst most candidates were able to correctly identify the median, the IQR appeared to be an unknown term or was found using 75 – 25 = 50 $\Rightarrow$ median = 3. A small minority of candidates even based their answers on the cumulative values of $\sum fx$. – In part (a)(ii), answers, usually directly from calculators, were in the main correct, although incorrect answers due to working only with x-values or only with f-values were seen.

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Q16.

Whilst answers here revealed an improvement in general standard over those to similar questions on previous papers, there is still room for improvement. Most candidates stated the correct value for the median but fewer were able to determine the IQR as 4 – 1 = 3. It was not unusual to see 285.75 – 95.25 = 190.5 $\Rightarrow$ IQR = 2. Correct values for the mean and standard deviation usually resulted directly from calculators but a minority of candidates stated, for example, $\bar{x} = \frac{380}{10} = 38$ revealing a total lack of understanding of both the data and the context. In answering part (b), where marks depended on approximately correct relevant answers to part (a), too many candidates failed to state overall conclusions for comparisons of average and spread as was requested. Simply comparing their four answers from part (a) with those of Jole and Katie was not sufficient.

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In answering part (b)(i), where marks depended on approximately correct relevant answers to part (a), too many candidates failed to state overall conclusions for comparisons of average and spread, as was requested. Simply comparing their four answers from part (a) with those of Jole and Katie was not sufficient. Even partially correct answers to part (b)(ii) were rare. Most candidates, making an attempt, usually based their answers on the validity of 95%, 2 or Katie's results. Hence it appeared that candidates were generally unaware of the basis for the statement. Having said this, a small minority of candidates were aware that Jole's statement was based on a normal distribution and that the data for 2005/06 was skewed.

Q18.

Despite this type of question appearing on recent papers, the ability of far too many candidates to correctly identify the median and quartiles from a list of values remains disappointing. In part (a), some candidates failed to even order the values, whilst other quoted the positions in the list as their values for the median and quartiles or used the runs scored as frequencies. Nevertheless, many candidates obtained 40 for the median and 50 for the interquartile range or at least 13 and 63 for the quartiles. In part (b)(i), most candidates stated incorrectly that no value occurred more than once whilst others, having stated that the mode was 0 , merely repeated the words of the question. Some candidates argued incorrectly that, since *a* was unknown, the mode could not be found. Answers to part (b)(ii) were better, with most candidates stating correctly that, since *a* or the largest value was unknown, the range could not be calculated.

Q19.

Almost all candidates scored the first 3 marks by use of the appropriate functions on their calculators. The small minority who used formulae were also generally successful. Scoring full marks in part (b) was much rarer. Most candidates made sensible attempts but often struggled to find the appropriate terms to use. Some candidates stated values for the upper and lower quartiles but did not then state the difference as their value for the interquartile range. A small minority of candidates first input 3 values greater than 60 into their calculators to then find correct answers. In part (c), general disadvantages of the mode and range were not acceptable answers. Whilst many candidates noted that no time occurred more than once so a mode did not exist, many fewer made explicit reference to the fact that the maximum time needed to calculate the range was unknown.

Q20.

The majority of candidates were apparently unprepared for this type of question. Whilst part (b) might be considered quite challenging to the average candidate, it was expected that most candidates would be able to make significant progress in part (a). This expectation was certainly not realised as only minimal marks were often awarded. Indeed it was not that rare to award a candidate only 1 mark for the whole question; this for correctly stating the modal value although, even here, an answer of 24 was not that unusual. Answers stated for the range were often 0 – 15 or 24 – 4 = 20.

In part (a)(ii), many candidates attempted interpolation, presumably on the basis that the data were continuous. Those who recognised its discrete nature rarely helped themselves by constructing a cumulative frequency table. However they were often able to identify the median as 3. Many candidates had less success with the quartiles. Even some of those who identified 2 and 4 did not take the difference to find the IQR whilst a small minority stated that the IQR = 72 – 24 = 48 or that this implied the value of 3.

In part (a)(iii), it was rare to see 7 and 12 identified as the two group mid-points, although from the considerable number of stated correct answers, direct from calculators, they had been used. Candidates who used formulae often failed to identify the frequencies and so took n as 8 or even 15. Even those candidates who found the correct value for the mean sometimes had insurmountable difficulties in finding the standard deviation.

Correct answers to part (b) were extremely rare. Candidates often chose to explain in detail what each statistic measured, or stated for example: 'median and IQR are not affected by extreme values or outliers'; 'median and IQR are whole numbers'; 'mean and standard deviation are closer together than mode and range'. Centres are encouraged to refer to a copy of the published Mark Scheme for acceptable answers.

Q21.

Some candidates did not identify the correct mid-points usually through being 0.5 too high. These usually led to an incorrect mean value but to a correct value for the standard deviation. A minority of very weak candidates simply took the frequencies as 8 values of x.

Q22.

This question proved a good source of marks for many candidates with the more able often gaining at least 15 marks. Answers to part (a)(i) were almost always correct and usually by use of the formula. In part (a)(ii), most candidates realised that 7.5 $\left(\frac{15}{2}\right)$ required $P(K \le 7)$ or $P(K < 8)$ and so scored full marks from tables. A small minority of candidates used P (K = 7) or $\frac{P(K \le 7) + P(K \le 8)}{2}$. As in previous papers, part (a)(iii) continued to cause more difficulties almost always through use of $P(K \le 3)$ and/or $P(K < 7)$. Knowledge of (an available source of) the relevant formulae for part (b)(i) was much improved with many fully correct answers as a result. When marks were lost, it was invariably for failing to find $\sqrt{3.6}$ for the standard deviation. Not surprisingly, only the very weakest candidates failed to score the 2 marks in part (b)(ii). In part (b)(iii), able and well-prepared candidates compared the two means (equal) and two standard deviations (different) but then often failed to commit to the overall conclusion of 'doubtful validity'. A minority of candidates only compared $\frac{\bar{x}}{15} = \frac{6}{15}$ = 0.4 with the value quoted (1 15 15 mark) or presented a qualitative argument as to the benefits of coaching (0 marks).

Q23.

Answers to this straightforward question were generally poor with many candidates scoring no marks. There was about a 50:50 split between those candidates using functions on their calculators and those using formulae. Common errors were to ignore the frequencies and so opt for n = 9, interchange frequencies (f) and values (x) or calculate $\sum(fx)^2$ instead of $\sum fx^2$.

Q24.

Statistics 1A

Candidates were often able to score at least some of the first 4 marks using the statistical functions on their calculators, particularly as the mark scheme allowed for some 'errors' in finding the class mid-points that were not integers. Candidates using formulae were often much less successful usually through working only with either class mid-points or class frequencies. When attempted by the more able candidates, answers to part (b) often referred to skewness or non-symmetry, although statements referring to the discrete nature of the data or likely negative values were equally valid. However statements on the size of the standard deviation were deemed invalid. Most seen answers to part (c)(i) were correct through mention of the Central Limit Theorem, or large sample size, although in many scripts it appeared that this was perhaps as good a place as any to state it. Answers to part (c)(ii) were much less impressive with typical common mistakes equivalent to mean = μ = 25.2 and variance = 17.2 or 17.2^2. Despite this, many candidates recovered to make a valid attempt at part (d) by using a correct expression with a correct z-value. Here the final accuracy mark also required accurate answers to part (a). Finally, in part (e), some candidates failed to notice that the first claim referred to 'average' and so 30 required a comparison with their confidence interval. To score the marks, this comparison had to be clear. A number of other candidates simply compared 30 with their sample mean. However, a surprising number of candidates answered the second claim correctly by identifying that only 7% of the sample values exceeded 50; in some cases these were the only marks scored on the question.

Statistics 1B

Candidates were often able to score the first 4 marks using the statistical functions on their calculators, particularly as the mark scheme allowed for some 'errors' in finding the class mid-points that were not integers. Candidates using formulae were often much less successful, usually through working only with either class mid-points or class frequencies. When attempted by the more able candidates, answers to part (b) often referred to skewness or non symmetry, although statements referring to the discrete nature of the data or likely negative values were equally valid. However statements on the size of the standard deviation were invalid. Most answers seen to part (c)(i) were correct through mention of the Central Limit Theorem or large sample size. Answers to part (c)(ii) were much less impressive with typical common mistakes equivalent to mean = μ = 25.2 and variance = 17.2 or 17.2^2. Despite this, many candidates recovered to make a valid attempt at part (d) by using a correct expression with a correct z-value. Here the final accuracy mark also required accurate answers to part (a). Finally, in part (e), some candidates failed to notice that the first claim referred to 'average' and so 30 required a comparison with their confidence interval. A number of other candidates simply compared 30 with their sample mean. However, a pleasantly surprising number of candidates answered the second claim correctly by identifying that only 7% of the sample values exceeded 50. A minority of candidates compared 50 with their confidence interval.

Q25.

Candidates generally found part (a) of this question straightforward. A mistake that was made fairly frequently was omitting to divide the sum of the squares by 10 before subtracting the square of the mean. There was some confusion between the variance and the standard deviation here. Part (b) was not well answered, many candidates finding the new value for the mean successfully but then not knowing how to modify $\sum x^2$ in order to recalculate the variance.

Q26.

In this question, the estimated mean was often found correctly but the candidates' attempts at the estimated standard deviation were riddled with errors and misconceptions.

Q27.

Part (a) was usually answered well. Those who were successful were generally able to obtain the mean in part (b) although the standard deviation in °C proved more problematic. Many candidates assumed that the two standard deviations must be equal.

Part (c) was not answered well. Frequently, candidates would perform an operation on one side of the equation (such as multiplying by 9) and then do something entirely different on the other side. Even those who coped numerically often reversed the inequality sign incorrectly.

Q28.

The part of this question that caused candidates most trouble in this question was finding the variance in part (a), although many candidates knew what to do and did it efficiently. A-high proportion of candidates knew how to cope with the scaling in part (b), some losing a mark by omitting to give units in their answers. In part (c), most candidates could modify the mean correctly but added one unit to the standard deviation as well.

Q29.

A surprisingly large proportion of candidates were unable to carry out the calculation in part (b). It was rare for candidates who failed at this stage to score many marks later in the question.

Q30.

Part (a) was well answered but part (b) was not. A common error was to confuse the standard deviation with the variance but many omitted this part completely. Part (c)(i) provided two easy marks for those candidates who used their calculators, but many candidates carried out lengthy and often incorrect calculations.

In part (c)(ii), many candidates ignored the instruction to relate the answer to part (c)(i) and answered in general terms instead. It was common for candidates to interpret the data in part (c) as the number of minutes Siballi had to wait for a bus rather than the number of times out of ten Siballi had to wait more than 6 minutes.

Q31.

On previous MBS1 papers questions on diagrams have been the least well answered. However, nearly all candidates this time were able to deal with line diagrams. There was some improvement compared to previous papers in finding the mean and standard deviation of a grouped frequency distribution in part (a), although there were still a substantial number of candidates who failed to find the class midpoints correctly.